WORKED SOLUTION

STEP I 2018, Question 6

Pure • Midpoint and Trapezium Rules

STEP I 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q6

The key idea

Both approximations are sums of sines of equally spaced angles, and the given product-to-sum identity makes such sums telescope once they are multiplied by a well-chosen $2\sin\theta$. The midpoint rule samples at odd multiples of $\frac{\pi}{2n}$ and the trapezium rule at even multiples, so with $\theta = \frac{\pi}{2n}$ each part becomes a telescoping sum. Once A_n and B_n are in closed form, part (iii) is just half-angle formulae.

Solution

The preliminary identity

Take $P = \theta$ and $Q = (2k-1)\theta$ in the identity:

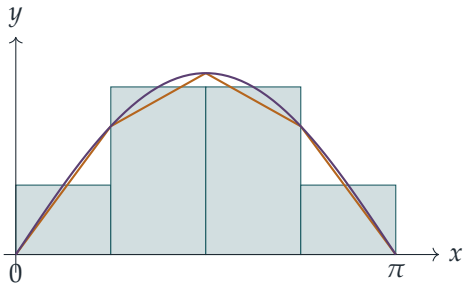
$$2\sin\theta\,\sin(2k-1)\theta = \cos(2k-2)\theta - \cos 2k\theta.$$

Summing from $k = 1$ to $k = n$, each $\cos 2k\theta$ subtracted in one term is added back in the next, so the sum telescopes:

$$\begin{aligned} 2\sin\theta\sum_{k=1}^{n}\sin(2k-1)\theta &= (\cos 0 - \cos 2\theta) + (\cos 2\theta - \cos 4\theta) + \cdots \\ &\quad + (\cos(2n-2)\theta - \cos 2n\theta) \\ &= 1 - \cos 2n\theta, \end{aligned}$$

as required.

Throughout the rest, write $h = \frac{\pi}{n}$ for the strip width and $\theta = \frac{\pi}{2n} = \frac{h}{2}$.



For $n = 4$: the midpoint rectangles (shaded) overestimate the area, and the trapezium chords (bold) underestimate it, because $\sin x$ is concave on $[0, \pi]$.

Part (i)

The kth strip runs from $(k-1)h$ to kh, so its midpoint is at $x = (k - \frac{1}{2})h = (2k-1)\theta$, and its height is $\sin(2k-1)\theta$. Hence

$$A_n = h \sum_{k=1}^{n} \sin(2k-1)\theta.$$

Multiplying by $2\sin\theta$ and using the identity, with $2n\theta = \pi$,

$$2\sin\theta\, A_n = h\,(1 - \cos\pi) = 2h.$$

Dividing by 2 gives $A_n \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{n}$, as required.

Part (ii)

The ordinates are at $x = kh = 2k\theta$ for $k = 0, 1, \ldots, n$, and the end ordinates are $\sin 0 = \sin\pi = 0$. So the trapezium rule gives

$$B_n = \frac{h}{2}\left(\sin 0 + 2\sum_{k=1}^{n-1} \sin 2k\theta + \sin\pi\right) = h \sum_{k=1}^{n-1} \sin 2k\theta.$$

These are now even multiples of θ, so we use the identity once more, this time with $P = \theta$, $Q = 2k\theta$:

$$2\sin\theta\, \sin 2k\theta = \cos(2k-1)\theta - \cos(2k+1)\theta.$$

Summing from $k = 1$ to $n-1$ again telescopes:

$$2\sin\theta \sum_{k=1}^{n-1} \sin 2k\theta = \cos\theta - \cos(2n-1)\theta.$$

? **Ask yourself** How do we simplify $\cos(2n-1)\theta$?

Since $2n\theta = \pi$, we have $(2n-1)\theta = \pi - \theta$, and $\cos(\pi - \theta) = -\cos\theta$. So the right-hand side is $2\cos\theta$, and

$$2\sin\theta\, B_n = 2h\cos\theta, \qquad \text{that is,} \qquad B_n \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{n}\cos\left(\frac{\pi}{2n}\right).$$

(When $n = 1$ the sum is empty and $B_1 = 0$; the formula still holds, since $\cos\frac{\pi}{2} = 0$.)

Part (iii)

Since $0 < \theta \leq \frac{\pi}{2}$, $\sin\theta \neq 0$, so parts (i) and (ii) give

$$A_n = \frac{h}{\sin\theta}, \qquad B_n = \frac{h\cos\theta}{\sin\theta}.$$

Replacing n by $2n$ halves both h and θ:

$$A_{2n} = \frac{h/2}{\sin(\theta/2)}, \qquad B_{2n} = \frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)}.$$

For the first result, use $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$:

$$\tfrac{1}{2}(A_n + B_n) = \frac{h}{2}\cdot\frac{1+\cos\theta}{\sin\theta} = \frac{h}{2}\cdot\frac{2\cos^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)} = \frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)} = B_{2n}.$$

For the second, using the same double-angle formula for $\sin\theta$ (and $\cos\frac{\theta}{2} \neq 0$),

$$A_n B_{2n} = \frac{h}{2\sin(\theta/2)\cos(\theta/2)}\cdot\frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)} = \frac{h^2}{4\sin^2(\theta/2)} = \left(\frac{h/2}{\sin(\theta/2)}\right)^2 = A_{2n}^2.$$

Both results are shown.

Checks

- $n = 1$: one midpoint rectangle of width π and height $\sin\frac{\pi}{2} = 1$ gives $A_1 = \pi$, matching $\pi/\sin\frac{\pi}{2}$; and $B_1 = 0$ as noted.
- $n = 2$: $A_2 = \frac{\pi}{2}(\sin\frac{\pi}{4} + \sin\frac{3\pi}{4}) = \frac{\pi}{\sqrt{2}} \approx 2.2214$ and $B_2 = \frac{\pi}{2} \approx 1.5708$; then $\frac{1}{2}(A_1 + B_1) = \frac{\pi}{2} = B_2$ and $A_1 B_2 = \frac{\pi^2}{2} = A_2^2$.
- $n = 10$: summing directly gives $A_{10} \approx 2.00825$ and $B_{10} \approx 1.98352$, either side of the true area 2. As $n \to \infty$, $\frac{\pi}{n}\big/\sin\frac{\pi}{2n} \to 2$ and $\cos\frac{\pi}{2n} \to 1$, so both tend to 2, as they should. All four results were also confirmed by computer for every n up to 200.

Common slips

- Putting the midpoints at kh rather than $(k - \frac{1}{2})h$, which turns the midpoint rule into a sum at the trapezium nodes.
- Forgetting that the trapezium rule halves the end ordinates. Here they vanish, which is why B_n is simply h times the interior ordinates.
- Leaving $\cos(2n-1)\theta$ unsimplified, rather than using $(2n-1)\theta = \pi - \theta$.

Beyond STEP

Part (iii) says that B_{2n} is the arithmetic mean of A_n and B_n, and A_{2n} is the geometric mean of A_n and B_{2n}. Taking reciprocals turns this into the harmonic-then-geometric mean pattern of Archimedes' method of squeezing π between polygons around and inside a circle, and the iteration is known as Borchardt's algorithm. More generally, comparing the midpoint and trapezium errors (the midpoint error is about half the trapezium error, with opposite sign) leads to Simpson's rule and to Richardson extrapolation, which are central ideas in numerical analysis. For the error analysis behind these rules, see Bartle and Sherbert, *Introduction to Real Analysis*, on the Riemann integral.

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